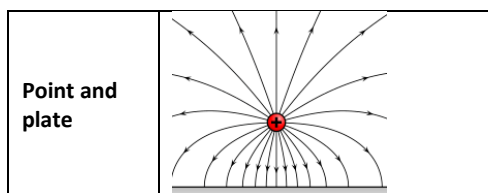


Quantities and units	
E	Electric field strength, unit Vm^{-1} or NC^{-1} .
	This is easily confused with energy! To avoid this confusion:
	Electric field strength is a vector, energy is a scalar. It is best to use W as the symbol for energy in this topic (think of it as work done; energy transferred = work done). Unfortunately, in the textbook E is sometimes used for energy (see p 439) and section 5 of this sheet.
V	Electric potential, unit V (volt)
Q, q	Charge, unit C (coulomb). Q and q represent two charges. Think of Q as the charge causing the field, and q the usually smaller charge experiencing the field. In some books Q_1 and Q_2 are used instead. They are in fact interchangeable – it is important to remember which is which when you are solving problems
k	In coulomb's law and elsewhere we have $4\pi\epsilon_0$. It is sometimes convenient to use k, where $k = \frac{1}{4\pi\epsilon_0}$. For example you could look up ϵ_0 in the data booklet and calculate k, to avoid repeatedly making the same calculation and save time. I advise against, as there is a risk of confusing $4\pi\epsilon_0$ with $\frac{1}{4\pi\epsilon_0}$



1. Electric Fields	
Electric field or electrostatic field	A region of space in which a charged object experiences a force.
Electric field strength	The force experienced, per unit positive charge. NC^{-1} $E = F/Q$ A vector quantity, the same direction as the force: the direction the positive charge would move
Electric field patterns: lines show the direction of the force on a positive charge, so point away from positive charges and towards negative ones . They meet surfaces at right angles	
Point charge	
Uniformly charged sphere	
Parallel plates	
Two points, same sign	
Two points, opposite sign	

2. Coulomb's Law	
Equation: inverse square law	$F = \frac{Qq}{4\pi\epsilon_0 r^2}$ Also written as $F = \frac{k Qq}{r^2}$ where $k = \frac{1}{4\pi\epsilon_0}$

3. Uniform E fields and capacitance	
In a uniform field	$E = \frac{V}{d}$
Units	E can have units Vm^{-1} as well as NC^{-1}
Capacitor	A pair of parallel plates with a material between them called a dielectric. The electric field will be as in the diagram for parallel plates
Dielectric	Insulating material between the plates of a capacitor (could be air, paper, or other materials) to increase the capacitance
Capacitance	$C = \epsilon \frac{A}{d}$ Where $\epsilon = \epsilon_r \epsilon_0$ ϵ_r is the permittivity of the dielectric ϵ_0 is the permittivity of free space
Isolated sphere	$C = 4\pi\epsilon_0 R$ Where R is the radius of the sphere

4. Charged particles in uniform electric fields	
A charged particle in an electric field will follow Newton's laws of motion The acceleration can be found using $a = F/m$ $F = EQ$ $E = V/d$	
Work done	$W = Fd$ $W = VQ$ ($V = W/Q$, definition of V)
Projectile motion	A charged particle in an electric field can be treated just like a projectile moving in a uniform gravitational field, with horizontal and vertical components treated independently

5. Electric potential energy	
Electric potential energy	$E = \frac{Qq}{4\pi\epsilon_0 r}$ This is <i>energy</i> , (unit J), not electric field strength
Electric potential	"The electric potential at a point is defined as the work done per moving charge in moving a positive charge from infinity to that point." $V = \frac{Q}{4\pi\epsilon_0 r}$ Compare this with the equation above: they are closely related. $E = Vq$ (or $W = Vq$)
Equipotential	A line along which the electric potential is the same. (Like a contour line, along which the height is the same). The equipotentials around a point charge would be concentric circles. In 3 dimensions we would have an equipotential surface.